ContractLens MAS — Comparative Synthesis Report

Broad Activity Schedule and first payment assessment under Option A and Option C. Specifi

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Agents: GCC Agent · NEC Agent · Comparison Agent
Workflow: Query Decomposition → Parallel Querying → Cross-examination → Synthesis → QA
Overall verdict: **PASSED**

Context / Scenario

A contract is awarded using ECC Option A for a new multi-storey car park. The Client left tenderers free to define activities on the Activity Schedule. A first-time tenderer identifies a document in Contract Data part two that looks like a bill of quantities summary, including broad items for steel reinforced concrete works and exterior blue steel cladding.

Topic

Broad Activity Schedule and first payment assessment under Option A and Option C.

Questions

How would your answer to part b. differ if it was an Option C contract?

Query Plan

L1 — Spirit / Philosophy: How does the payment regime allocate cash-flow risk and incentivise transparency where the pricing document is broad or insufficiently detailed?
L2 — Clause-level Provisions: What clauses govern the first payment assessment, including the amount due, assessment dates, activity completion, Price for Work Done to Date, Defined Cost, Fee, and any distinction between Option A and Option C?
L3 — Practice Implications: What practical approach should the certifier or project manager take when assessing early payment under a broad Activity Schedule, and how would that approach change for an Option C target cost contract?
Canonical topic: payment_mechanism

Anchor Clauses

GCC: —
NEC4: cl. 11.2(24), cl. 11.2(29), cl. 11.2(31)

Comparative Synthesis

> **Executive summary —** GCC has no Activity Schedule regime: first payment is BQ/Schedule-of-Rates valuation. NEC Option A bars payment for incomplete broad activities, while Option C pays Defined Cost plus Fee. This shifts practice from Engineer-led measurement to proactive activity design or open-book cost evidence.

1. Spirit / Philosophy — Measured valuation vs open-book governance.

GCC: The GCC payment ethos is Engineer-led, valuation-based, and claims-neutral for this topic: cash flow follows measured work and Contract rates, not activity completion or cost reimbursement. In TCE terms, it uses hierarchical certification and itemised pricing to reduce valuation uncertainty, with the Engineer controlling

dissatisfaction and minimum-payment filters.

NEC: NEC4 ECC is process-based and relational, requiring the Project Manager to assess at each assessment date in a spirit of mutual trust and co-operation under cl. 10.2. Option A deliberately transfers cash-flow risk to the Contractor where activities are broad, while Option C reduces that risk through transparent Defined Cost plus Fee assessment; in TCE terms, NEC substitutes structured information exchange and auditability for purely ex post valuation.

2. GCC Position (Implementation) — BQ valuation governs first payment.

Implementation flow:

1. Contractor submits interim statement at end of each monthly period, first commencing on date for commencement of the Works (Clause 78(1)) →
2. Statement shows work value, Site materials, and further sums due (Clause 78(1)) →
3. Engineer values and certifies sum due within 21 days of receiving the statement (Clause 79(1)) →
4. Engineer may omit dissatisfied work, materials, or services (Clause 79(5)) →
5. Engineer may refuse certificate if assessed sum is below the Contract minimum (Clause 79(2)).

Clause 59(1) is cited within the GCC payment mechanism, but the expert response stresses that GCC has no equivalent terminology for “Activity Schedule”, “Price for Work Done to Date”, “target cost arrangement”, “Defined Cost”, “Fee”, or “Disallowed Cost”. Financial valuation is governed by Bills of Quantities or a Schedule of Rates, not by an activity-based payment schedule or target-cost mechanism.

Under Clause 78(1), the Contractor must submit an interim statement “at the end of each monthly period”, with the first monthly period “commencing on the date for commencement of the Works”. The first statement must show: “the estimated contract value of the work done in accordance with the Contract up to the end of such monthly period”; “a list of materials delivered to the Site for use in the permanent work and their estimated contract value”; and all further estimated sums the Contractor considers due.

Under Clause 79(1), the Engineer must value and certify the sum due “within 21 days of receiving the statement”. The Engineer assesses the value “based on the rates in the Contract where appropriate”, including permanent work executed, any Temporary Works or preliminary items provided for in the BQ, and materials properly stored on Site. Because payment follows actual physical work executed against individual BQ rates, a high-level activity being incomplete does not prevent payment for measured work.

The Engineer’s certification powers are material. Clause 79(5) gives a broad power to “omit from any certificate the value of any work done, materials supplied or services rendered with which the Engineer may for the time being be dissatisfied”. Clause 79(2) allows the Engineer to refuse an interim payment certificate if the total assessed sum is less than the minimum payment stated in the Contract. The GCC is therefore silent on forecast or actual cost evidence under a Defined Cost plus Fee target-cost arrangement.

3. NEC4 Position (Implementation) — Option A milestones; Option C cost.

Implementation flow:

1. Contractor submits payment application before the assessment date, detailing assessment (cl. 50.2) →
2. Project Manager assesses amount due at the assessment date, acting under cl. 10.2 and cl. 50.1 →
3. Amount due includes PWDD plus other amounts, less retentions or Contractor payments (cl. 50.3) →
4. Option A: PM pays only completed activities/groups in PWDD (cl. 11.2(29), cl. 11.2(32)) →
5. Option C: PM assesses Defined Cost forecast before next assessment date plus Fee, with records and Disallowed Cost checks (cl. 11.2(31), cl. 11.2(24), cl. 52.2, cl. 11.2(26)) →

6. PM certifies payment within one week of the assessment date (cl. 51.1).

The Project Manager assesses the amount due at each assessment date, acting in a spirit of mutual trust and co-operation under cl. 10.2. The Contractor must submit an application for payment before the assessment date, detailing how the amount has been assessed, under cl. 50.2. The amount due includes the Price for Work Done to Date, plus other amounts, less amounts to be retained or paid by the Contractor, under cl. 50.3. The Activity Schedule is strictly a pricing document and is not Scope or Site Information under cl. 55.1.

Under Option A, the Prices are the lump sum prices for each activity on the Activity Schedule under cl. 11.2(32). PWDD is the total of the Prices for "each group of completed activities and each completed activity which is not in a group" under cl. 11.2(29). A completed activity is one without Defects which would delay or be covered by immediately following work. If an activity is only 90% complete, its PWDD value is zero. Where broad activities stifle cash flow, the Contractor may change its planned method of working and submit a revised Activity Schedule breaking work into smaller billable activities under cl. 55.3. The PM accepts if the changed Prices are reasonably distributed and the total of the Prices is not changed under cl. 55.4; the PM replies within the period for reply.

Under Option C, the answer differs fundamentally. The Activity Schedule establishes the target cost, namely the total of the Prices for the Contractor's share assessment under cl. 54.1, but is not used to calculate interim payments. PWDD is instead the "total Defined Cost which the Project Manager forecasts will have been paid by the Contractor before the next assessment date plus the Fee" under cl. 11.2(31). Defined Cost is the cost of components in the Schedule of Cost Components less Disallowed Cost under cl. 11.2(24). The Contractor must keep detailed accounts and records of payments of Defined Cost and proof that payments have been made under cl. 52.2. The PM assesses actual and forecasted cost evidence and deducts costs "not justified by the Contractor's accounts and records" as Disallowed Cost under cl. 11.2(26).

4. Behavioural Implications — Cash flow depends on proof structure.

Under GCC, the Contractor's payment behaviour is monthly and valuation-led: prepare an interim statement, quantify physical work done, list Site materials for permanent work, and state further estimated sums. The Engineer then measures, values by BQ/Schedule-of-Rates logic, applies dissatisfaction and minimum-payment controls, and certifies within 21 days. The incentive is to maintain reliable measurement and Site-material records, but not to restructure work into payment milestones.

Under NEC Option A, cash-flow risk is front-loaded into the Activity Schedule. The Contractor must apply before the assessment date and prove completed activities or completed groups; near-complete work does not earn PWDD if the activity remains incomplete. The practical incentive is to design the Activity Schedule at tender or revise it under cl. 55.3 into smaller, reasonably distributed billable activities without changing the total Prices.

Under NEC Option C, the behavioural focus changes from completion milestones to open-book cost substantiation. The Contractor must maintain accounts, payment records, proof of payment, and forecasts of Defined Cost plus Fee. The PM's role is less about measuring completed activities and more about checking actual and forecasted cost evidence, deducting unsupported items as Disallowed Cost. Broad activity drafting is therefore a cash-flow problem under Option A, but mainly a target/share-accounting issue under Option C.

Implementation comparison (GCC vs NEC4)

Aspect	GCC	NEC4
Pricing basis	BQ/Schedule of Rates valuation; Clause 79(1).	Activity Schedule Prices or Defined Cost; cl. 11.2(29), cl. 11.2(31).

Application timing	End of each monthly period; Clause 78(1).	Before the assessment date; cl. 50.2.
Certifier timing	Engineer certifies within 21 days; Clause 79(1).	PM certifies within one week; cl. 51.1.
Broad schedule effect	No Activity Schedule concept; measured work still valued.	Option A gives zero for incomplete activities; cl. 11.2(29).
Option C distinction	No target cost, Defined Cost, Fee, or Disallowed Cost.	Activity Schedule sets target, not interim payment; cl. 54.1.
Evidence focus	Work quantities and Site materials; Clause 78(1).	Accounts, payment proof, forecasts; cl. 52.2.

5. GCC Lessons for NEC Adoption — GCC routines become NEC controls.

GCC practitioners already know the discipline of monthly interim statements under Clause 78(1): identify work done, Site materials for permanent work, and further sums due. That discipline transfers to NEC cl. 50.2 payment applications, but the behavioural adjustment is timing and structure: applications must be ready before the assessment date and must explain the NEC basis of assessment, not merely measured value.

The GCC habit of valuing physical work against BQ or Schedule-of-Rates items under Clause 79(1) supports NEC Option A only if it is converted into activity-completion control. Instead of expecting payment for partly completed measured work, practitioners must ensure the Activity Schedule is divided into smaller billable activities, or revise it under cl. 55.3 for PM acceptance under cl. 55.4, with changed Prices reasonably distributed and the total Prices unchanged.

GCC record routines for materials, quantities, and valuation checks also transfer to NEC Option C, but their purpose changes. Under cl. 11.2(31), cl. 11.2(24), cl. 52.2, and cl. 11.2(26), the PM assesses Defined Cost plus Fee and deducts unsupported costs as Disallowed Cost. The practical adjustment is evidential: maintain accounts, payment records, proof of payment, and credible forecasts, rather than relying on completion of activities.

The practical reframing is simple: use GCC claim, record, variation, and delay-analysis habits earlier under NEC as activity-design, payment-application, Defined Cost substantiation, and cash-flow forecasting tools.

GCC practice / experience	Transfers to NEC implementation
Clause 78(1) monthly interim statement.	cl. 50.2 application; submit before assessment date with NEC basis.
Clause 79(1) BQ/Schedule-of-Rates valuation.	Option A cl. 11.2(29); convert measurement into completed activities.
Recording Site materials under Clause 78(1).	Support payment evidence; note HK Option A cl. 50.2A.
Engineer checks dissatisfaction under Clause 79(5).	Option C cl. 11.2(26); avoid unsupported Disallowed Cost.
Minimum-payment awareness under Clause 79(2).	Option A cl. 55.3; restructure activities to protect cash flow.

6. HK-Specific Notes — PAH and DEVB modify local practice.

GCC: The Project Administration Handbook, Chapter 9, explicitly maps contract terminology to prevent confusion. It establishes that GCC contracts use “Bills of Quantities / Schedule of Rates” rather than an “Activity

Schedule", and use "Contract Sum" rather than "Prices" or "Price for Work Done to Date". The GCC corpus is completely silent on target cost arrangements, Defined Cost, Fee, Disallowed Cost, and requirements for forecast or actual cost evidence in interim assessments.

PAH Chapter 9 also states that, under the Standard Method of Measurement for Civil Engineering Works, interim payments are assessed by recording quantities of each completed BQ item in an Interim Payments Record Book or through running abstract sheets. For lump sum contracts with firm BQ, the Engineer estimates the amount of work done, if possible in agreement with the Contractor, and values it based on the BQ rates. This confirms that GCC payment assessment is independent of broad activity completion rules, Defined Cost plus Fee mechanisms, and Disallowed Cost parameters.

NEC4 Hong Kong: The DEVB Practice Notes and NEC4 ECC HK Edition contain modifications addressing cash flow and administrative burdens at the first assessment. For Option A, DEVB cautions that "Unless there is an individually itemised activity in the Activity Schedule, part payment for an activity would not be made". Hong Kong public works projects frequently adopt a standard optional amendment, cl. 50.2A, allowing part payment for "Plant and Materials on site" before the activity is fully completed.

For Option C, the HK Edition amends the PWDD definition: cl. 11.2(31) / HK Edition cl. 11.2(34). Rather than purely forecasting, PWDD is assessed based on the actual payment made by the Contractor at the assessment date, plus an estimated amount for cost components that will be paid "up to one week after the next assessment date". This is intended to prevent "negative cash flow" where Contractors must pay costs before reimbursement.

DEVB also mandates that, for the first 6 payment applications under Option C, the Contractor must provide a full set of supporting documents for amounts already paid for full checking by the Project Manager. After these first six applications, a risk-based payment checking approach is adopted, using sample checking based on value and Degree of Irregularity to reduce administrative burden. Under the HK Edition library of standard amendments, the Project Manager certifies payment within two weeks, rather than one week, of the assessment date, and payment is made within three weeks.

Rule-based Validation Summary

Overall verdict: PASSED
Confidence: 1.00
Iterations used: 1
Rule-based track: pass

Metric	Value	Threshold	Status
Citation Accuracy Rate (GCC)	1.00	≥ 0.85	pass
Citation Accuracy Rate (NEC4)	1.00	≥ 0.85	pass
Coverage Rate (GCC)	1.00	≥ 0.85	pass
Coverage Rate (NEC4)	1.00	≥ 0.85	pass
Composite Confidence	1.00	≥ 0.70	pass

NEC4 anchors covered: cl. 11.2(24), cl. 11.2(29), cl. 11.2(31)

LLM Semantic Validation (Track 2)

Track verdict: fail
Severity: moderate
Default codes: E1 (hallucinated clause), E2 (egregious content mismatch)
Diagnostic-only — does NOT gate the overall verdict.

#	Type	Type description	Issue
1	E1	Hallucinated clause — cited in synthesis but absent from raw evidence	The synthesis cites NEC/HK Option A cl. 50.2A, but cl. 50.2A does not appear anywhere in the NEC raw evidence.
2	E1	Hallucinated clause — cited in synthesis but absent from raw evidence	The synthesis cites HK Edition cl. 11.2(34), but cl. 11.2(34) does not appear anywhere in the NEC raw evidence.

Proposed gap queries (for retry)

• Retrieve NEC4 ECC Hong Kong Edition or DEVB amendment evidence for cl. 50.2A and HK Edition cl. 11.2(34), including Option A part payment for Plant and Materials on site and Option C amended PWDD assessment.